



FINAL YEAR PROJECT REPORT

THEME :

**Enhancing embeddings in Youth Informal French through
Domain-Adaptive Fine-Tuning of French-Based Local NLP
Models (FLAUBERT CASE)**

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ABBREVIATIONS

AUC : Area Under the Curve

DAFT: Domain Adaptive Fine Tuning

LAS : Labeled Attachment Score

LLM : Large Language Model

MAE : Mean Absolute Error

MLM: Masked Language Modeling

MSE : Mean Squared Error

NER : Named Entity Recognition

NLI : Natural Language Inference

NLP : Natural Language Processing

NLU : Natural Language Understanding

RMSE : Root Mean Squared Error

ROC : Receiver Operating Characteristic

SLU : Spoken Language Understanding

SOTA : State-Of-The-Art

STS : Semantic Textual Similarity

Abstract

In today's fast-paced world where artificial intelligence is now establishing itself as one of the pillars of society, many innovations are emerging every day. The emergence of these models reveals a linguistic disparity, with French-based models outnumbered by English-based ones despite the strong French-speaking presence in the world and especially in the industry. To address this issue, BERT-like models have raised, reducing the existing gap.

However, it must be acknowledged that these models were essentially trained on formal data, limiting their ability to grasp the informal varieties that are omnipresent in discussions and interactions and, consequently, their comprehensive understanding of the French language.

In recent years, several studies, such as 88milSMS [1], Sedinkina [2], and Tremolo-Tweets [3], have attempted to address this issue by both building new datasets and exploring new methods for adapting to language variations, leading to major improvements in models such as CamemBERT, but these improvements are still insufficient given the scale of the problem.

This study was designed to determine to what extent it is possible to improve the ability of French-based models to process informal French by refining their embeddings and semantic representations. The results show an improvement in Spearman and Pearson correlations from 0.3200 and 0.2378 to 0.8216 and 0.8056 respectively compared to the baseline model, suggesting a significant improvement.

1. Introduction

To reduce the linguistic gap between French and English resulting from the emergence of artificial intelligence models in recent years, BERT-like models [4] have been developed, representing a significant advance in NLP. From automatic proofreaders to virtual assistants, their pre-trained approach has increased the accuracy of language models in several tasks such as classification and translation. Based on NLU, they have facilitated a more detailed understanding of the meaning of language elements in their respective contexts by the models. However, despite the impact of their contribution, it must be acknowledged that they still have a major flaw: informality. Indeed, apart from written elements such as speeches, letters or articles, communication in French is largely punctuated by informality, which is found in many conversations, whether between humans or between humans and machines. As a result, these models face a major limitation in that they are unable to process informal French, as they have been trained on formal data such as articles, newspapers, Wikipedia, etc. They are therefore currently unable to grasp all the linguistic variations of the French language.

In the literature, several studies such as 88milSMS [1] and Tremolo-Tweets [3] have attempted to address this problem by building dedicated corpora or developing adaptation methods, but despite the relevance of their contributions, solutions still need to be found.

This gives rise to the main issue of this study, which is **to determine how can these models be adapted to informal language to improve their semantic representations and embeddings?**

The answer to this question will be based on various areas of study, namely:

- *How can the size and nature of corpora influence the ability of models to handle informal French?*
- *What are the major challenges and limitations associated with improving the ability of these models to capture the lexicon and semantics of informal French?*
- *What training method is the most effective for improving the quality' vector representations?*

This research aims to significantly enhance the semantic representation of informal

French within an established French model. In the following lines, this study will evaluate the model chosen to assess the performance and limitations of French LLMs on informal language, propose ways to improve vector representations, and then compare the results obtained with baseline performance to prove the relevance of the study.

2. Literature Review

2.1. Introduction

NLP is a discipline whose goal is to enable machines to understand and interact with the language that we humans use every day. It is the basis for many of the innovations that have emerged over the last ten years and is key to their effectiveness and accuracy.

All this is thanks to embeddings, which represent words as numerical vectors that can be manipulated by machines, and language models (LLMS), which form the backbone of modern text comprehension and generation systems.

However, there are still several limitations at the heart of these technologies, particularly when it comes to informal language. Texts such as slang, abbreviations and verlan do not follow conventional rules, making them difficult to analyse. This is particularly noticeable in the case of the English language, as pre-trained models based on English have been developed from formal sources, thus limiting their ability to handle informality.

The aim of this literature review is to study existing scientific work on this subject, highlighting its contributions and shortcomings.

2.2. Large Language Models

According to Wikipedia (1), Large Language Models (LLM) are neural network models capable of understanding, manipulating and generating text from natural language. In other words, they are large models that are trained on huge quantities of data (text) from different sources so that, like human beings, they can grasp the structures, relationships and meanings of words in each language. Their operation is based on the Transformer architecture and the use of embeddings for text encoding. (2).

They are used in many everyday applications such as machine translation, chatbots and virtual assistants. The first language models appeared in the 1990s with the emergence of rule-based models such as ELIZA. The work of Weizenbaum et al, 1996 [5] consisted of dialogue

programmes based on predefined linguistic rules (what might also be called pattern matching). We then saw a move towards more learning models based on the probability of word sequences. These were n-grams and Hidden Markov Models (HMM). They respectively brought to the NLP field the ability to predict a word based on previous words according to the meaning of the sentence (which is what we see today when we write on our phones or computers) and the ability to capture the transition probabilities between states (the states being the “hidden” source of the words used). However, although the models had been learned, they still lacked a true understanding of meaning, due to the lack of a vector representation of meaning. In the 1990s, the field was revolutionised by the arrival of neural networks, with the introduction of technologies such as RNN (Recurrent Neural Network), LSTM (Long Short-Term Memory) and Seq2Seq (Sequence to Sequence). Their aim: to enable machines to understand and manipulate human language more fluidly and contextually. In this vein, RNNs were the first to emerge. There are several works on the subject, but one of the most relevant to the field of NLP is that of J. Elman (1990) [6], in which several concepts are revealed. It is now possible to process sequential data, in this case the sentences we use every day. Moreover, the way is now open to sequential processing in which previous states are kept in a hidden state to define the context (this is known as a “SIMPLE RECURRENT NETWORK”). However, RNNs have a major flaw: vanishing gradients, i.e. limited short-term memory. The data stored is quickly forgotten, which limits the capacity of these models to process long sentences. In response to the vanishing gradient problem, Hochreiter. S et al (1997) [7] proposed an optimal solution, LSTMs. The idea is to integrate long-term memory mechanisms to provide models with a larger and more powerful memory. This is made possible using a memory cell and gates (input and output) that control and manage the flow of information according to several mathematical formulae. However, LSTMs also have the problem of variable inputs and outputs. An LSTM on its own is unable to provide the output with everything it has had since the input if the data is too large. It tends to retain only the end according to its capacity and to process the sequences without often taking account of the sequences already processed, which often leads to repetition. It was only a few years later that, to overcome this problem, it was proposed to use two LSTMs, one to encode the input and the other to decode the output. This was the birth of the Seq 2 Seq architectures. SutSkever et al, (2014) [8] introduced an interesting solution through their work on the subject. That of using two LSTMs, the first to read the input and compress (summarise) it so that the architecture can retain it and a second at the output which will be able to decompress it to present it to the user. This is the model on which all machine translation and summarisation models, and even some chatbots, are based today. In this way,

they have paved the way and laid the foundations for more powerful and more efficient models, namely the Transformers. Although LSTM / Seq 2 Seq are efficient and important architectures. As technology advances, so do its requirements. Very quickly, these architectures became slow in relation to demand, offering average precision without forgetting that they still carried certain memory leaks. This led to the emergence of Transformer architectures. These are the foundations behind models such as CHATGPT, BERT and so on... Introduced by Vaswani, A et al, (2017) [9] through their work they introduce a whole new concept. Instead of reading words sequentially, they propose to read all words in parallel and use an attention system to understand the relationships between them all based on embeddings, self-attention layers and feed-forward networks. Based on the Transformer models, concrete and practical applications have emerged: Large Language Models.

2.3. Challenge of informal Language Processing in French

In recent years, studies and research have paved the way for machines to better understand natural human language. Today, most emerging models are multilingual, allowing a wider audience to benefit from them. However, this multilingual capability limits their efficiency in each language. This is demonstrated by the work of Martin L. et al, (2020) [10], in which he shows the extent to which a monolingual model can outperform multilingual models for a specific language. Indeed, after performing four similar tasks, namely morphosyntactic tagging, syntactic analysis, named entity recognition and logical inference on both datasets, it was found that a 4 GB monolingual dataset outperformed a 130 GB multilingual corpus. In line with this, this research will be monolingual, based on French.

French-based LLMs are trained on primarily formal sources, which limits the models' ability to cover informal variations such as slang, abbreviations, etc. This is illustrated by Lam T. et al. (2024) [11] through CamemBERTv2, an improved version of CamemBert [10] with more robustness, in which they attempted to introduce noisy and informal data into the corpus. The results showed that the model was able to achieve an F1 score of up to 97% for formal English, fell below 80% for informal English. Consequently, even if the diversity of data in the pre-training corpus was increased, this did not really improve the models, as the lexical innovations identified in text messages or slang remain difficult to process using the established rules of

linguistic models.

Magueresse et al (2023) [12] also showed in their study that linguistic distortions such as slang or abbreviations are poorly understood by models, even after attempts at adaptation. Indeed, after drawing data from sources such as youth forums, text messages and social networks such as Twitter, and after testing models such as BERT/CamemBERT, it appears that processing informal language results in a loss of 8 to 20 F1 score points in terms of classification and entity detection.

The work of Dementieva, D et al, (2023) [13] highlighted the problems associated with register bias. After a comparative study between formal and informal language in terms of classification tasks on three models, BERT, mBERT and XLM-R, and two corpora, one formal consisting of professional emails and the other consisting of text messages; The results show that the models make three times more errors in detecting informal French, with an error rate of 6.6% for informal French compared to 18.7% for formal French. In addition, the mBERT model achieved an F1 score of 72% for informal language, compared to 84% for formal language. Thus, the study highlights that the models tend to produce many false positives, mainly due to confusion between context and meaning.

In short, informal French remains a complex area that LLMs are still unable to master due to several factors, including lexical complexity and register bias. In the following lines, the work undertaken to address this problem will be examined.

2.4. Existing works on French Language Models

To address the issue of understanding informal language, various methods such as fine tuning, data augmentation, and the creation of new datasets are being implemented.

This section will examine some recent relevant work, analysing their contributions and drawbacks.

First, Pillet, B. et al, (2021) [14] present TremoLo-Tweets, a study mainly focused on fine-tuning and data augmentation. TREMoLo-Tweets is a refined version of CamemBERT, trained

on a sub-corpus of tweets manually annotated according to three categories. The data is then automatically augmented with highly probable labels, increasing the size of the dataset to approximately 228,505 tweets without excessive human cost. The results show an improvement leading to an F1 score of approximately 83% after data augmentation, then 85% after fine-tuning, compared to 95% for the formal model, thus reducing the significant gap with the initial model. Several major conclusions emerge, notably the fact that fine-tuning and data augmentation do indeed improve the processing of informal data, even if only slightly. In addition, certain limitations should be noted, including the model's inefficiency on data other than that from Twitter, the poor quality of self-annotations, and the limited coverage of unnamed slang.

Shortly before, Cougnon et al. (2013) [15] also explored other improvement solutions, including the creation of a dedicated corpus and the increase in data to address the existing data gap. The proposed study '88milSMS' highlights the creation of a new annotated dataset to help models better capture language diversity, particularly in an informal context. The results show that this has led to gains of +7 to +15 points in F1 or precision on sentiment detection tasks. However, the disadvantage is that the data can only be used in registers close to the source dataset, which complicates its generalisation. Also, despite the high cost of its creation, it does not cover all informal French, suggesting the need for further improvements.

Furthermore, Sedinkina, M. (2021) [16] introduces 'Domain Adaptation' in her thesis, which consists of adapting sub-parts of the LLM to a new domain without having to work on the entire LLM itself. This allows the foundations and semantics of the models to be retained while teaching them new variations. After performing classification and detection tasks, the results showed a gain of +5 to +10 points in terms of accuracy or F1 score, thereby significantly reducing the decline in model performance at a lower cost.

Nevertheless, due to its complex configuration, this method is not within everyone's reach, and moreover, it becomes obsolete and requires updating to handle new informality's variations.

In conclusion, many methods have been and continue to be implemented, including those listed above and many others (vector post-processing, adversarial training, prompt-based adaptation, and stylistic adaptation).

This study will take the existing limitations of these solutions as a basis for proposing an innovative and effective solution.

3. Methodology and Research Design

3.1. Research Design

3.1.1. Type of research

This study is secondary research, as it relies on secondary data. To build the dataset that will be used to refine the chosen LLM, data will be extracted from various existing and published datasets. It will then be sorted according to relevance and compiled to form the study's dataset. In addition, the methodology used will be based on existing methods that will be adapted and improved to obtain better results.

This research is also mixed, mainly qualitative but also quantitative.

It is qualitative because the aim of the study is to improve the LLM's ability to understand informal variations in French, so observations, analyses and interpretations will be necessary for a better understanding of the resolution process.

The quantitative aspect is reflected in the use of numerical metrics for model evaluation, and in the performance comparison that will determine the relevance of the study.

3.1.2. Ethical considerations

This study aims to respect the international community and web users. To this end, it is based on data from public sources intended for community use, and therefore no individual consent is required. Furthermore, the data has been double-checked to certify that no personal information is used, and it will also be stored securely so that it can only be used for this research. In line with this, all proper names that may be found in the dataset have been replaced by the acronym 'Anon' to preserve anonymity and confidentiality. Furthermore, the use of the dataset complies with current academic regulations, university policy and terms of use.

3.1.3. Timescale

This study was conducted using an agile methodology. Its main sprints are corresponding to key phases in the development of the study over a twelve-week period. Among these, the construction of the corpus and the data augmentation process were carried out over a four-week period. Data cleaning and preparation took one week. The application of fine-tuning methods and the adjustment of hyperparameters to achieve optimal results took place over a three-week period. The evaluation and comparison of the performance of the basic model and the improved model, as well as the final adjustments for the acquisition of the results, took two weeks. Finally, the documentation was also written over a two-week period.

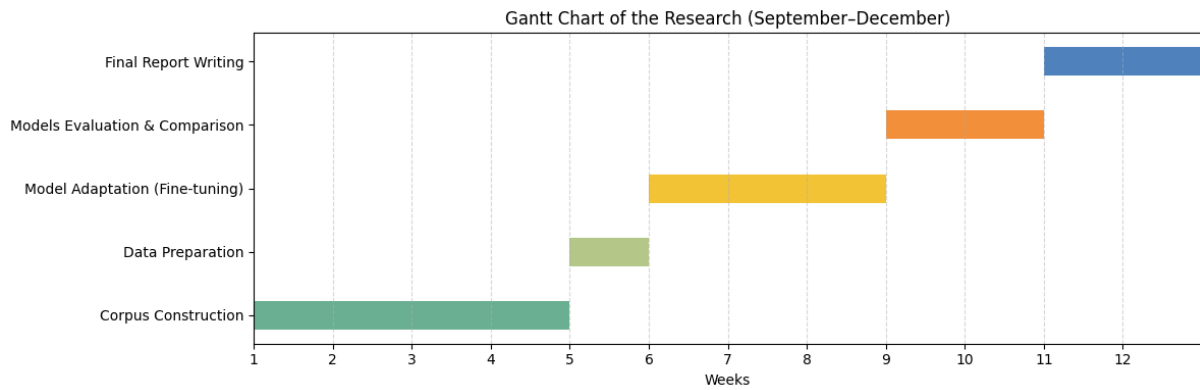


Figure 1 : Gantt Chart of the Research

3.2. Methodology Design

3.2.1. Chosen Language Model

FlauBERT [17] was selected as the base model for this study. Flaubert is a Transformer-type LLM pre-trained primarily on large French text corpora. Its official documentation states that out of approximately 71 GB of data and nearly 13 billion tokens, it contains only about 2 GB and 226 million informal tokens, or less than 3% of the total corpus, making it a model consisting of 97% formal data, ideal for this study, unlike models such as Camembert [18], whose proportions are unknown. Inspired by Google's BERT model and with an architecture similar to that of XLM-RoBERTa[19], it has demonstrated SOTA performance on tasks such as classification, NER and many others, demonstrating its robustness in capturing complex syntactic variations in the French language, positioning it as one of the best benchmarks in the field. Flaubert also presents itself as a better solution compared to multilingual models such as Mistral [20] or LLaMA [21], primarily because of its monolingual nature, which positions it as a better choice for working on a specific chosen language [15]. Indeed, unlike multilingual models, which are optimised for versatility rather than informality, it is better suited to the context. Furthermore, unlike multilingual models, which do not allow for complete fine-tuning without heavy infrastructure and are sometimes too large for the context, Flaubert does not require a lot of resources. Finally, it is an academic and open-source benchmark commonly used as a basis in industry through applications such as machine translators, sentiment detectors, chatbots, or in more specific fields such as banking or law.

However, when it comes to performance in informal contexts, Flaubert ranks among the weakest in this area. According to the study conducted by Oralie Cattan et al.(2022) [22], where a comparison of the performances of some French LLMS on an SLU task using datasets such as MEDIA [23] or ATIS-FR [24], (mainly composed of spontaneous discussions reflecting informal nuances of language,) has been made, the result reveal that Flaubert_base performs the worst, with a CER of 8.1, making it the ideal model to improve for this research.

Model	Dataset	F1 Score	CER
FlauBERT base	MEDIA	89.0	8.1
CamemBERT base	MEDIA	89.9	7.5
FrALBERT base	MEDIA	89.8	8.6
XLNet base	ATIS-FR	92.5	5.3

Table 1 : LLMs performances comparison on SLU task [22]

Consequently, the FlauBERT configuration that will be used is the FlauBERT_base_cased [25] variant, a Transformer Encoder Only architecture whose specifications are listed in (Table 2). This version is lighter and faster for fine-tuning while offering a stable and complete architecture, unlike larger models such as FlauBERT_large. Furthermore, unlike the uncased version, this version offers the ability to better capture the nuances of language. In the informal language the way of writing determines intonation or emotion, therefore this variant offers the case management necessary for the study. It makes the difference between upper- and lower-case letters, which is beneficial for better management of context and language nuances.

Parameters	Specifications
Architecture type	Transformer Encoder Only
Layers	12
Hidden Size	768
Attention heads	12

Table 2 : Table of specifications for FlauBERT_base_cased parameters

3.2.2. Data and Corpus

According to an article published by Touro University [26], there are four main forms of informal language: contractions, phrasal verbs, slang/colloquialisms, and first-person pronouns. Therefore, in order to be as representative as possible of informal French, this study will focus on the “slang/colloquialism” type, which includes the most variations outside the register compared to the others.

The dataset used for this study will mainly consist of elements such as slang, abbreviations, and swear words, (the sources of which will be specified in the following lines) to best represent the informal variations of the French language.

To carry out this study, two datasets will be set up:

- The main dataset: contains the informal sentences and elements that will be used to train and adapt the model.
- The STS dataset (second dataset): Separate from the main dataset, this dataset contains pairs of annotated informal sentences that will be used to evaluate the baseline (the performance of the basic model) and the performance of the model after fine-tuning.

Furthermore, the processes of data collection, augmentation and preparation will be detailed below.

3.2.2.1. *Data Collection*

A. Main Dataset

The main corpus for this study was constructed from data from four other open-source corpora: 88milsms [15], Jigsaw Multilingual Swear Dataset [27], ProfanityFilter_datasets [28] and A Corpus of 1000 Authentic SMS in French [29].

These datasets were chosen because of their relevance to this study, as they contain swear words (in the case of Jigsaw Multilingual Swear Dataset [27], ProfanityFilter_datasets [28]) that are widely used in informal discussions, and elements from real telephone conversations, responses from users on social networks and youth forums (88milsms[15] and A Corpus of 1000 Authentic SMS in French [29]) that reflect variations in language in a real and continuous context.

This category of data was adopted over others, such as the sole use of tweets, because words and expressions take on meaning in context, something that is often lacking when dealing solely with tweets.

The data acquisition process targeted lexical elements such as slang, abbreviations and interjections, as well as syntactic elements such as deliberate errors and spoken grammatical structures. In addition, data acquisition was also carried out using approximately 80 keywords and informal expressions (e.g. ‘mdr’, ‘pk’, ‘tfk’, ‘svp’, ‘tkt’) to obtain the most relevant results possible.

In total, the main dataset is structured in a single column named “text” and consists of 1,013 items distributed according to the description given in *Table 3*.

Dataset Name	Source	Size	Type	Description/ Purpose	Proportion in the main dataset
88milSMS	http://88milsms.huma-num.fr	88000 elements	Text	Corpus of real telephone conversations via text message	54%
Jigsaw Multilingual Swear Dataset	https://www.kaggle.com/datasets/mikelgr500/jigsaw-swear-dataset	435755 elements	Text	Multilingual dataset consisting of rude	3%

	multilingual-swear-profanity			expressions and swear words.	
ProfanityFilter_datasets	https://github.com/devYRPauli/ProfanityFilter/blob/main/datasets/french.csv	95 elements	Text	Set of coarse and/or taboo French words	8.5%
A Corpus of 1000 Authentic SMS	https://doi.org/10.18167/DVN1/0ZGJRC	1000 elements	Text	Corpus of 1,000 messages enriched with spatial information.	34.5%

Table 3: Datasets Description

After the data augmentation process, this corpus will be divided as follows:

- **Training (80%):** for training the model
- **Validation (10%):** for validating and adjusting hyperparameters during training
- **Testing (10%):** for internal evaluation of the model before using the STS dataset

B- STS dataset

The STS_dataset is a dataset designed for STS fine tuning and model evaluation during the STS task. It is a very important dataset in this research because it serves as a reference for the model in understanding semantics and as a key element in performance evaluation.

It consists of data sourced 50% from the main dataset and 50% from the 88milSMS corpus, extracted according to the same criteria used for acquiring elements from the main dataset. It contains 500 pairs of informal elements, each followed by a similarity score structured in three separate columns: “sentence1”, “sentence2” and “similarity_score”.

Furthermore, the process of annotating the pairs, i.e. assigning the similarity score, will be explained in the following lines.

C-Pairs annotation process

Annotating pairs consists of giving them a score based on their semantic similarity. To do this, the study uses the Gold Standard [30], a standard benchmark for evaluating semantic similarity between pairs of sentences, to annotate the standard STS_dataset according to the following criteria:

- 0: independent meaning, no semantic relationship
- 5: Same meaning, semantic equivalence
- [1-4]: Degrees of equivalence according to contextual meaning, nature of words, etc.

This annotation system also allows for floating point numbers with up to 4 decimal places to best capture the nuances of language.

However, regarding the human and manual nature of this method, it becomes inappropriate when annotating large datasets such as those obtained after data augmentation.

Therefore, to annotate the pairs in the STS dataset after data augmentation, another method was used: bootstrapping, as mentioned and used in the study by Gao T. et al. (2021) [31]. This method consists of three main phases, which are as follows:

- Using the annotated corpus of 500 pairs to fine-tune the base model to obtain a v0 model.

- Using this v0 model to annotate the remaining thousands of pairs. Through a specific algorithm, the v0 model assigns scores to the new pairs based on the cosine similarity it has integrated via the 500 annotated pairs used to tune v0.
- Using the final annotated dataset for the rest of the study.

3.2.2.2. *Data preprocessing*

This research pre-processing process aims to preserve as much of the raw data as possible. In contrast to traditional methods that focus on data standardisation, this method aims to preserve informal linguistics in its pure essence.

A. Data Augmentation

i. Main dataset

The data augmentation process of the main corpus is based on two main methods which were implemented: **the acquisition of additional data and the data transformation.**

Augmentation through data acquisition:

To enrich the main dataset, two data acquisition methods were implemented:

- Sampling of 88milSMS: From the large 88milSMS dataset, sampling was carried out, leading to the acquisition of **49,739 elements**. To do this, an extraction process based on keywords such as ['tt', 'ouai', 'wsh', 'dsl', 'ds', 'mec', 'flemme', 'chier', 'lsp', 'bsx', 'nn'] was implemented.
- Data extraction from Twitter: Twitter's Tweepy API and the use of hashtags such as [wsh, azy, flemme, belec, nan, or slangFR] via a specific algorithm enabled the acquisition of a total of **92 additional elements**.

Augmentation through data transformation:

The expansion of the corpus through data transformation is based on nine different techniques, as detailed below:

- Synonyms Replacement: The use of the **NLTK library** made it possible to use dictionaries that also contain informal language, such as **WordNet and OMW-1.4**. These dictionaries enabled words to be replaced by their synonyms in sentences, introducing new lexical variations while preserving the original meaning of the sentences. In total, this method generated around **4600 additional elements**.

- Sentence Shuffling: This method involves randomly restructuring sentences to increase syntactic variation. In total, it generates approximately **9460 elements**.

- Noise Injection: This involves adding characters/symbols that do not belong to the linguistic register to simulate informal environments, which are mostly noisy. In total, this method allows for the acquisition of **4500 additional elements**.

- Word Dropout: This technique involves randomly removing elements from sentences to encourage the model to recover the context of the sentence even from incomplete data. It generates approximately **9269 elements**.

- Word Elongation: This technique involves creating a repetition of letters within the same word. It is often used to simulate spoken grammatical syntax transcribed in writing to reflect emotions. In total, it allows **1010 elements** to be acquired.

- Spelling error: This involves simulating spelling mistakes that appear very often in informal contexts (intentionally or unintentionally) so that the model can understand them and link them to real words. In total, this method generates **1011 elements**.

- Contextual augmentation: This technique relies on the use of **DistilBERT** via the **nplaug library** [32] to generate augmented sentences while ensuring that the semantics and context of the sentences are preserved. In total, it allows for the acquisition of **6066 elements**.

- Word insertion: This involves randomly inserting predefined informal words and expressions (e.g. oseille, jpp, bats les couilles) into sentences to increase the informal vocabulary of the corpus. This method generates **9099 elements**.

-Sentence Split: This method consists of splitting long sentences into shorter elements to create new fragments and new syntax. In total, it generates **6,728 elements**.

Augmentation technique	Volume generated
Sampling of 88milSMS	+49,739
Data extraction from Twitter	+92
Synonyms Replacement	+4600
Sentence Shuffling	+9460
Noise Injection	+4500
Word Dropout	+9269
Word Elongation	+1010
Spelling error	+1011
Contextual augmentation	+6066
Word insertion	+9099
Sentence Split	+6,728

Table 4 : Summary table of the data augmentation process of the main dataset

ii. STS dataset

The process of augmenting the STS dataset revolves around three main techniques.

-Random Pairing: This technique involves generating random pairs from the sources listed above, adding variety to the corpus. It has generated a total of 5,000 new pairs.

-Hard Negative Mining: This involves creating pairs that are lexically close but semantically distant to help the model rely not only on the lexicon to establish correlation, but also on semantics. This method generated a total of 2,000 new pairs.

-Back translation: This technique involves translating sentences into another language and then retranslating them into English. In this study, the sentences were translated into English and then back into English, adding formal correspondences to the informal dataset, thereby refining the model's ability to understand informal language. This generated 2,000 new pairs.

Augmentation technique	Volume generated
Random Pairing	+5000
Hard Negative Mining	+2000
Back translation	+2000

Table 5: Summary table of the STS dataset data augmentation process

iii. Conclusion

In summary, after the data augmentation process, the main dataset received 101,574 new data points, increasing it from 1,011 elements to 101,574 elements. The STS dataset was enriched with 9,000 new pairs, increasing it from 503 to 9,503 pairs.

Though these datasets are already good and ready for use, they need a process of cleansing and filtering to be the most suitable for this study. Therefore, these processes will be detailed in the following lines of this report.

Dataset	Size before Data Augmentation	Size After Data Augmentation	Volume added
Main Dataset	1011	102585	+101574
STS Dataset	503	9503	+9000

Table 6 : General summary table of data augmentation

B. Data Encoding

The datasets used for this study are all **CSV** (.csv) files encoded in **UTF-8** to ensure that special characters are understood and that they are compatible with the Transformer architecture. The separator used is the comma (comma delimited).

C. Data Cleansing

i. Deletion

To ensure the best quality for the main and the STS datasets, the data was cleaned to keep only the most relevant elements. The following elements have been removed:

- Empty elements
- Duplicates: Exact duplicates and inverse duplicates (in the STS dataset) were deleted to ensure no redundancy.
- Data containing only numbers or symbols.
- Emojis and emoji-like punctuation: Despite their frequent use, they were not deemed necessary for this study and were therefore removed.
- Formal elements: Words and expressions identified as too formal based on the dictionary were also removed to retain as many informal elements as possible.
- Tags, URLs, and any other elements of a similar nature have also been removed

- Names: All proper names in the datasets were removed and replaced with ‘Anon’ to ensure the ethic side of the study.
- Asymmetric pairs: Pairs where a sentence is longer than the other have been removed. It has been made based on a **length ratio greater than 5:1** to ensure the consistency of the associated data.

ii. Conservation

To preserve the informal essence of the data, the following attributes have been retained:

- Punctuation: repeated punctuation where the number of symbols is between 1 and 5 has been preserved. In informal contexts, punctuation is an important marker of intonation and emotional emphasis, which is why it is important to preserve it.
- Sentence length: unlike standard studies, which generally remove sentences of fewer than 3 words for better standardisation, this study preserves even elements containing only a single word. As informal language is full of abbreviations and monosyllables, removing elements of fewer than 3 words would result in a significant loss of relevant data.

iii. Conclusion

After completing the data cleaning process, the main dataset now contains **100,585** elements and the STS dataset **9,427**. The results are shown in the following table :

Dataset	Size before Data Cleansing	Size After Data Augmentation
Main Dataset	102585	100585
STS Dataset	9503	9427

Table 7 : Comparative table of data volume before and after cleansing

D. Data standardisation

i. Tokenisation

As part of this study, no vectorisation or manual tokenisation was performed; the data was left in its raw format. The sentences were not converted into embeddings or vectors. This step is an integral part of the Transformers' action during training and therefore does not need to be performed during data pre-processing. Furthermore, as the FlauBERT model itself is equipped with a tokeniser, it will handle the conversion to IDs and data processing.

ii. Case sensitivity

The data was treated as case-sensitive, as writing a complete sentence in capital letters could indicate a different meaning from the same sentence written in lowercase letters. In informal contexts, capital letters often indicate a particular intonation or emotion that needs to be considered, so a distinction was made between upper- and lower-case letters.

iii. Truncation

Truncation and the creation of the attention mask will be done directly when the data is loaded by the Transformer.

EData Separation

The main dataset will be divided as follows:

- Training (80%) : for training the model
- Validation (10%): for validating and adjusting hyperparameters during training
- Testing (10%): for internal evaluation of the model before using the STS dataset

The STS dataset will be divided into two parts:

- Training (50%) : for STS fine tuning
- Testing (50%) : for evaluating model performance

F. Conclusion

In summary, the data collection and preparation process involves five main stages: collection, augmentation, cleaning, formalisation and division of data, all described in the following pipeline:

DATASET CONSTRUCTION PIPELINE



Figure 2 : Pipeline of the datasets creation process

3.2.3. Task Definition

For this study based on improving embeddings, the task chosen is **Semantic Textual Similarity (STS)**. This task aims to determine the degree to which embeddings capture semantic similarity. To do this, it uses a score ranging from 0 to 5 to quantify the degree of correspondence. It is often evaluated by metrics such as Spearman's correlation, Pearson's correlation, and many others that will be detailed later in this study.

3.2.4. Validation of Pairs

The gold score used for corpus annotation is based on the average of several human annotators' evaluations. Since these scores reflect prior agreement among annotators, they are considered reliable without requiring additional validation.

3.2. 5. Overall Methodological Approach

The main objective of the proposed approach is to evaluate the performance of French LLMs when faced with variations of the language that fall outside the official register, and then to improve them so that they can understand them easily. The idea is to take them away from their core domain to see how they react and then train them to master this new domain.

This study is therefore structured around seven main phases:

- Selection of the base model,
- Construction of the datasets,
- Evaluation of the base model's performance,
- Application of fine-tuning methods,
- Evaluation of the fine-tuned model's performance,
- Comparison of performance to prove the relevance of the study,
- Conclusion and writing of the final documentation.

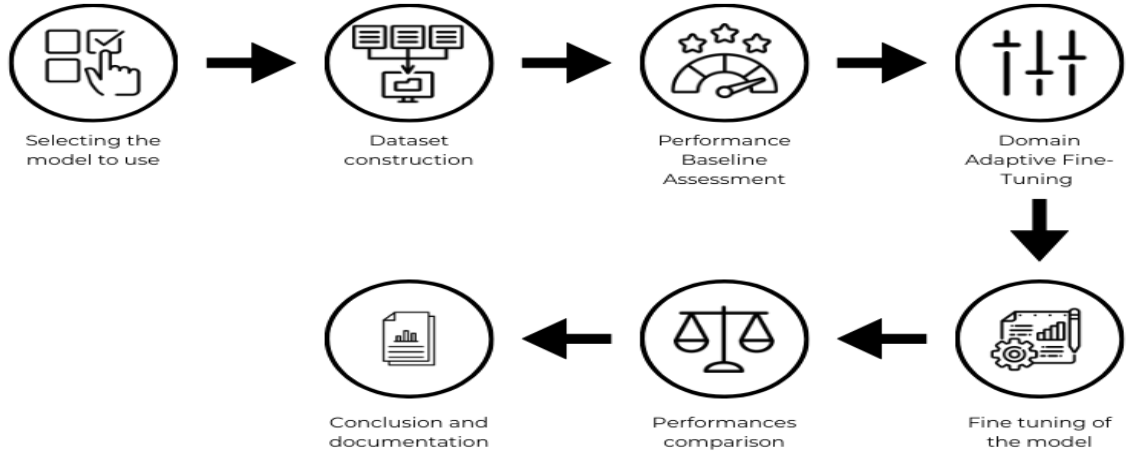


Figure 3 : Study methodology pipeline

This is also a supervised learning project, as the model learns from labelled data. The proposed methodology is an adaptation of the one used in the study by Qian Chen et al. (2023) [33], in which he introduces improvements to the quality of BERT embeddings using the same STS task to address the model's anisotropy problem. However, they did not use supervised learning, which resulted in a dependency on annotated data and reduced performance. This limitation is addressed in this study.

4. Experimental Setup

Improving embeddings for adaptation to the informal domain is a complex process involving several steps. The general idea was to perform DAFT, a technique that did not prove effective. Therefore, after DAFT, a second fine-tuning was performed on the regression head of the model, providing satisfactory results. The process, environment, and hyperparameters of the implementation are detailed below.

4.1. Hardware

This project was implemented on an HP Victus laptop running Windows 11 Home version 10.0.26200. This computer is equipped with a 12th Gen Intel Core i5-12500H processor, an Intel Iris Xe Graphics GPU and 16GB of RAM.

Data pre-processing and training were carried out using the software environment described in the following table:

Steps	Resources	Versions
Preprocessing	Visual Studio Code	1.106.3
	Python	3.13.2
	CUDA	11.8
	GPU	NVIDIA GEFORCE RTX 4060
	RAM	16GO
Training and Testing	Visual Studio Code	1.106.3
	Python	3.13.2
	CUDNN	9.1.0
	GPU	NVIDIA GEFORCE RTX 4060
	RAM	16GO

Table 8 : Table of material resources used

4.2. Tools and platforms

Various libraries were required to complete this project. Here is a detailed description of these libraries:

- Transformers (v 4.57.1) to load, fine-tune and evaluate the model. This is the library used to access the LLM.
- PyTorch (v2.9.1) to accelerate execution using the GPU (CUDA, CUDNN) to efficiently manage work on neural networks.
- Sentence-transformers (v5.1.2) to convert sentences into embeddings.
- Scikit-learn (1.7.2) which enables data processing and common machine learning tasks such as regression or classification.
- Pandas (v2.3.3) used for data set manipulation, from pre-processing to analysis
- Numpy (v2.3.1) used for complex mathematical calculations such as those required in the calculation of evaluation metrics
- Seaborn (v0.13.2) used for data visualisation
- Matplotlib (v3.10.3) also used for data visualisation via the creation of graphs or curves.
- Tensorflow (v2.9.1) used for deep learning of the model

4.3. DAFT

DAFT is a method that enables dynamic adaptation of LLM fine-tuning in the context of deep learning. DAFT aims to enhance model performance by adjusting their training parameters according to the data and the study objective. It addresses the issue of the traditional fine-tuning, which works on the entire model. DAFT allows it to work on specific layers or parts of the model for greater precision and flexibility in the adaptation process. It also allows for better use of hardware resources. It follows the pipeline below:

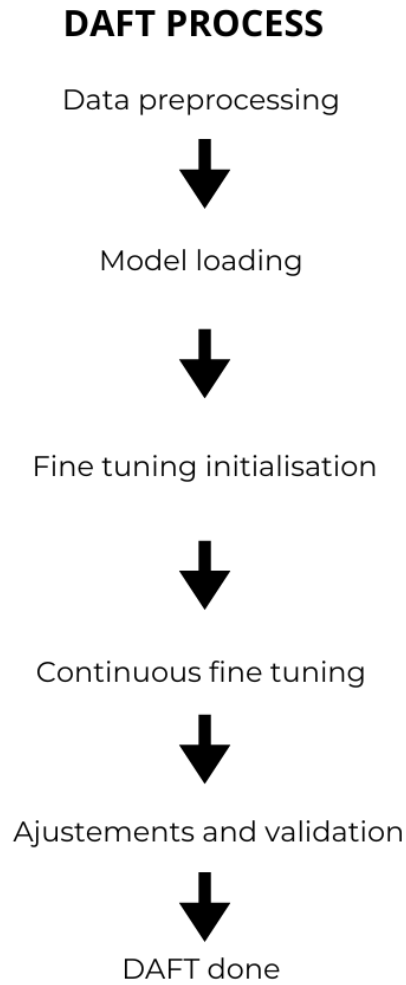


Figure 4 : Pipeline of the DAFT process

4.4. Fine tuning STS

Fine tuning is a process of adapting an LLM to a specific task.

In LLMS, the head refers to the output layer, which in most cases is used for predictions. STS fine-tuning therefore consists of working on this head, training it so that it determines perfectly and can predict the similarity between pairs of informal sentences. It takes place according to the following pipeline:

STS FINE TUNING PROCESS

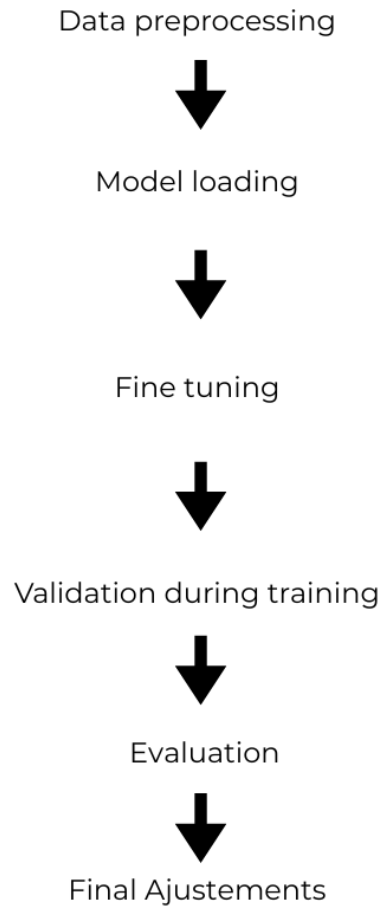


Figure 5 : Pipeline of the STS finetuning process

4.5. Hyperparameters and threshold adjustment

FlauBERT was adapted to informal language using a masked language modelling (MLM) task, which continuously pre-trains the model to learn new vocabulary and semantics from the main corpus. The hyperparameters used and the adjustment thresholds will be detailed below.

4.5.1. Hyperparameters

Tuning parameters is essential in the optimisation of the performance of LLMS such as FlauBERT. They allow the training hyperparameters to be modified to obtain the best possible results.

- Before adjustment, the main tuning parameters were as follows:
 - DAFT:
 - Masking rate: 15%
 - Train & eval batch size: 8
 - Epoch: 5
 - Steps: 500
 - Learning rate 5×10^{-5}
 - Weight decay: 0.01
 - Fine tuning STS:
 - Masking rate: 15%
 - Train & eval batch size: 8
 - Epoch: 5
 - Steps: 500
 - Learning rate 5×10^{-5}
 - Weight decay: 0.01
- After adjustment, the main tuning parameters are as follows:
 - DAFT:
 - Masking rate: 15%
 - Max_length: 128
 - Train & eval batch size: 8
 - Epoch: 15
 - Early_stopping: 3
 - Learning rate: 2×10^{-5}
 - Weight decay: 0.01

- Fine tuning STS:
 - Epoch = 8
 - Batch_size = 8
 - Num_labels = 1
 - Learning rate = 2×10^{-5}
 - AdamW with cosine annealing schedule

Adjusting the thresholds resulted in improved results. The initial thresholds were too small to be optimal and allow the model to learn effectively.

4.5.2. Threshold adjustment

Adjusting thresholds for models is crucial to optimising their performance in terms of accuracy, recall, and F1 score. Thresholds can refer to several aspects of processing or training.

Firstly, the learning rate was adjusted from 5×10^{-5} to 2×10^{-5} because a lower learning rate allows the model to converge more stably and learn the patterns present in the training data in a more refined manner.

In addition, the lower layers of the model have been frozen. This is because the lower layers (0-1-2) only learn the low level of language, in this case grammar, spelling and word representation. Only the upper layers learn semantic and contextual representations. By freezing these layers, we avoid erasing FlauBERT's basic knowledge of English while teaching it new variations.

4.6. Evaluation metrics

To evaluate the model's performance before and after fine-tuning, evaluation metrics are required. The metrics used in this research are:

- Regression metrics:

Their role is to evaluate the accuracy of the model in the establishment of similarity between the informal elements presented to it. These are:

- Spearman's correlation, which measures the linear relationship between predicted scores and actual scores.
- Pearson's correlation, which measures the monotonic relationship between predicted scores and actual scores.
- MSE (Mean Squared Error), which is the square mean of the errors.
- The Mean Absolute Error (MAE), which is the average of the absolute values of the errors.
- The Root Mean Squared Error (RMSE), which is the square root of the mean square of the values.
- The R^2 Score, which measures the variance of the actual score predictable from the model.

- Classification metrics:

For this study, similarity is binarised by adopting a threshold of 3.0

- Recall: Measures the proportion of similar pairs that the model has identified as similar.
- Precision: Measures the proportion of similar pairs that the model has predicted as similar.

- The F1 score is the harmonic mean of recall and precision.

5. Evaluations and results

5.1. Results

5.1.1. Baseline

The FlauBERT_base_cased model was evaluated before the start of the study to establish its baseline performance. The results are as follows:

- MSE: 14.4414,
- MAE: 3.4394,
- RMSE: 3.8002,
- R^2 : -4.5286,
- Spearman 0.3200
- Pearson: 0.2378

5.1.2. Results after DAFT

After performing DAFT on the FlauBERT model, a second evaluation was carried out. Here are the results.

- Spearman: -0.2798
- Pearson: -0.2361

5.1.3. Results after STS fine tuning

After performing STS fine tuning, a third evaluation was carried out. The results are as follows.

- Spearman: -0.2069

- Pearson: -0.1313

5.1.4. Final results

After STS fine tuning, AutoModelForSequenceClassification was used instead of AutoModel for direct use of the fine-tuned regression head. Here are the results:

- MSE: 0.05
- MAE: 0.0425
- RMSE: 0.226
- R² Score: 0.9
- Spearman: 0.8126
- Pearson: 0.8056

Steps	MSE	MAE	RMSE	R ² Score	Spearman	Pearson
Baseline	14.4414	3.4394	3.8002	-4.5286	03.200	0.2378
DAFT	N/A	N/A	N/A	N/A	-0.2798	-0.2361
STS FT	N/A	N/A	N/A	N/A	-0.2069	-0.1313
Final	0.05	0.0425	0.226	0.9	0.8126	0.8056

Table 9 : Summary table of model evaluation after each stage

5.2. Visualisation and interpretation of results

52.1. Evolution of evaluation metrics

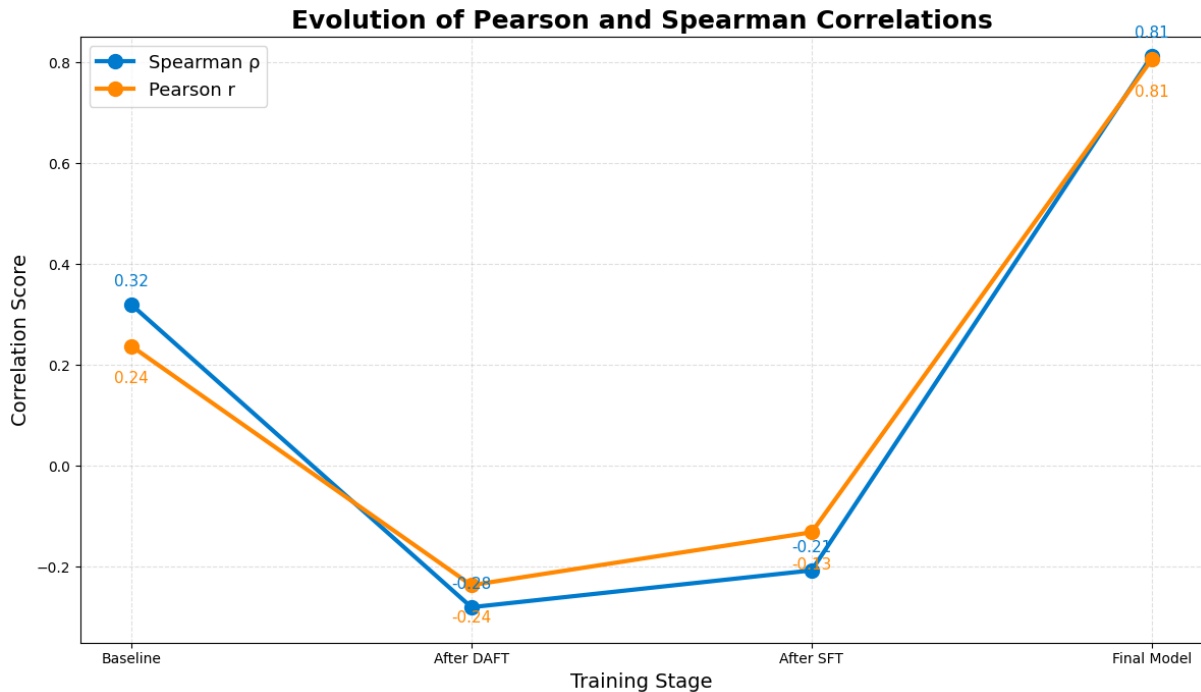


Figure 6 : Evolution of Spearman and Pearson Correlations

With the baseline established, the aim of the study is to exceed baseline performance.

After a DAFT operated with the main dataset, we can see a considerable drop in correlations. This is Catastrophic Forgetting. It can be explained by the fact that the model, having been so finely tuned to jargon, has lost its original semantics. The new embeddings have become so specific to informal language that they have broken the vector alignment. The model can recognise the vocabulary but still knows nothing about the relationships between the elements that make up that vocabulary. It also can be related to the amount of data used for the training. To address this problem, we perform fine-tuning of the head (STS fine-tuning). However, this does not yet solve the problem, because although performance has been improved, it is still not optimal. There is a slight improvement, but not enough to surpass the base model. This is due to the use of AutoModel, which leads to a manual calculation of cosine similarity and ignores the regression head that we trained by default.

Finally, when using `AutoModelForSequenceClassification` instead of `AutoModel`, i.e. the regression head, we see a satisfactory increase in metrics, surpassing the basic performance of the model and demonstrating the relevance of the study.

5.2.2. Confusion Matrix

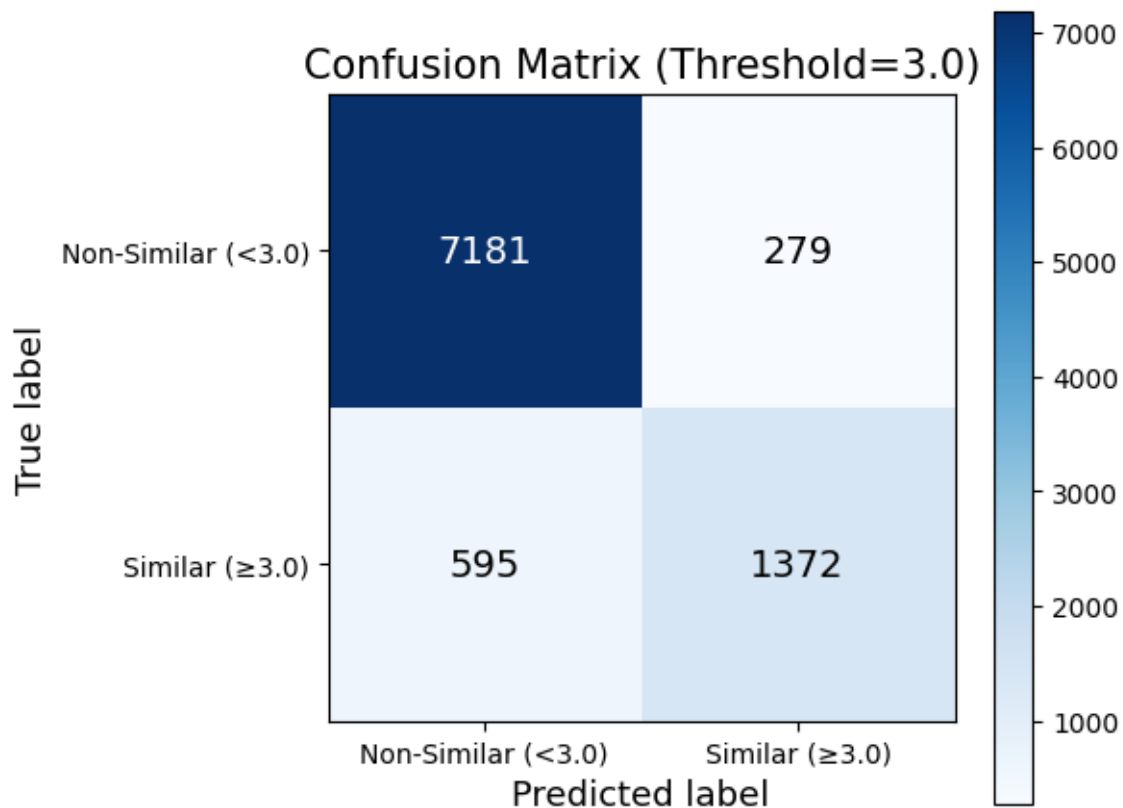


Figure 7 : Confusion Matrix

The confusion matrix shows a very high-performing model overall, combining 90.7% accuracy and 83.1% precision. It also reveals the existence of a conservative bias, demonstrating the model's sensitivity through 595 false negatives. The model is a little too cautious and may tend to incorrectly classify pairs as dissimilar.

The solution would be to adjust the binarisation threshold to make the model more confident.

5.2.3. ROC Curve

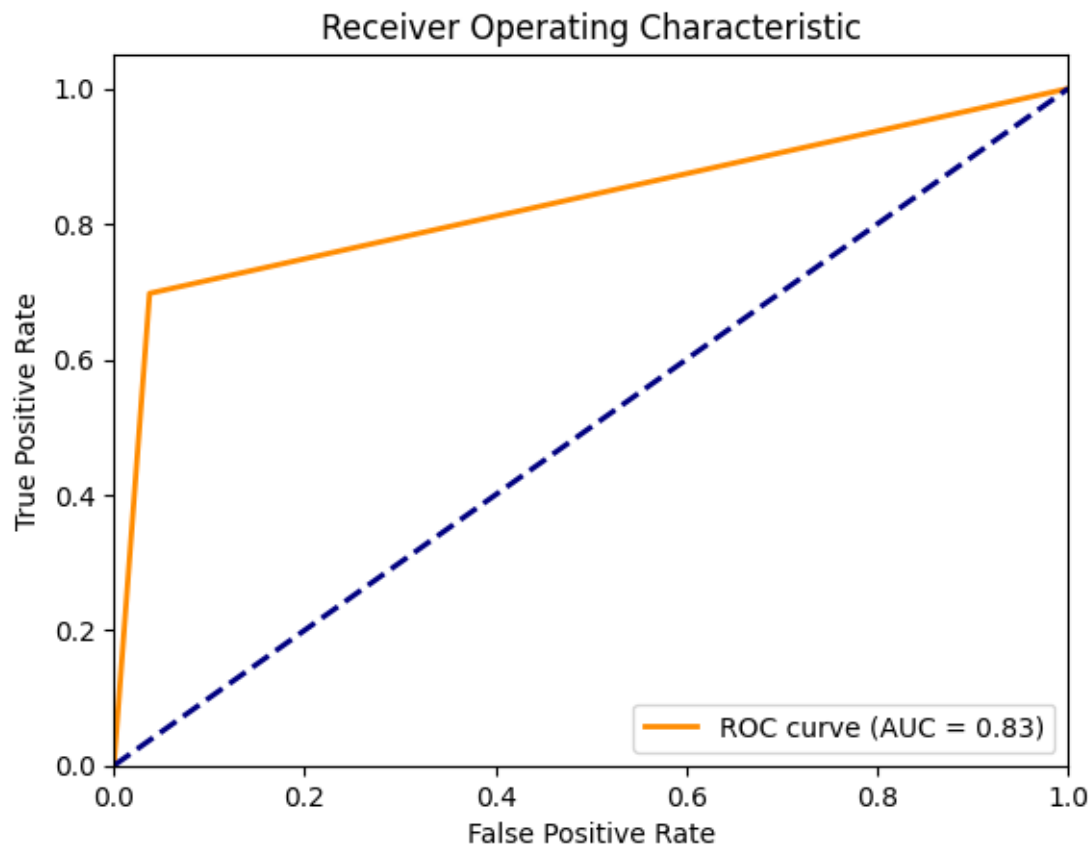


Figure 8: ROC Curve

The ROC curve measures the model's performance independently of the binarisation threshold. The AUC indicator is 0.83, which indicates that the model has an 83% chance of correctly classifying any given pair. Furthermore, the appearance of the curve shows a high trend on the left between 0.6 and 0.8, indicating that the model can determine a satisfactory rate of true positives without creating more false positives than necessary. In short, this attests to the robustness and efficiency of the model.

5.2.4. Distribution of Cosine Similarities

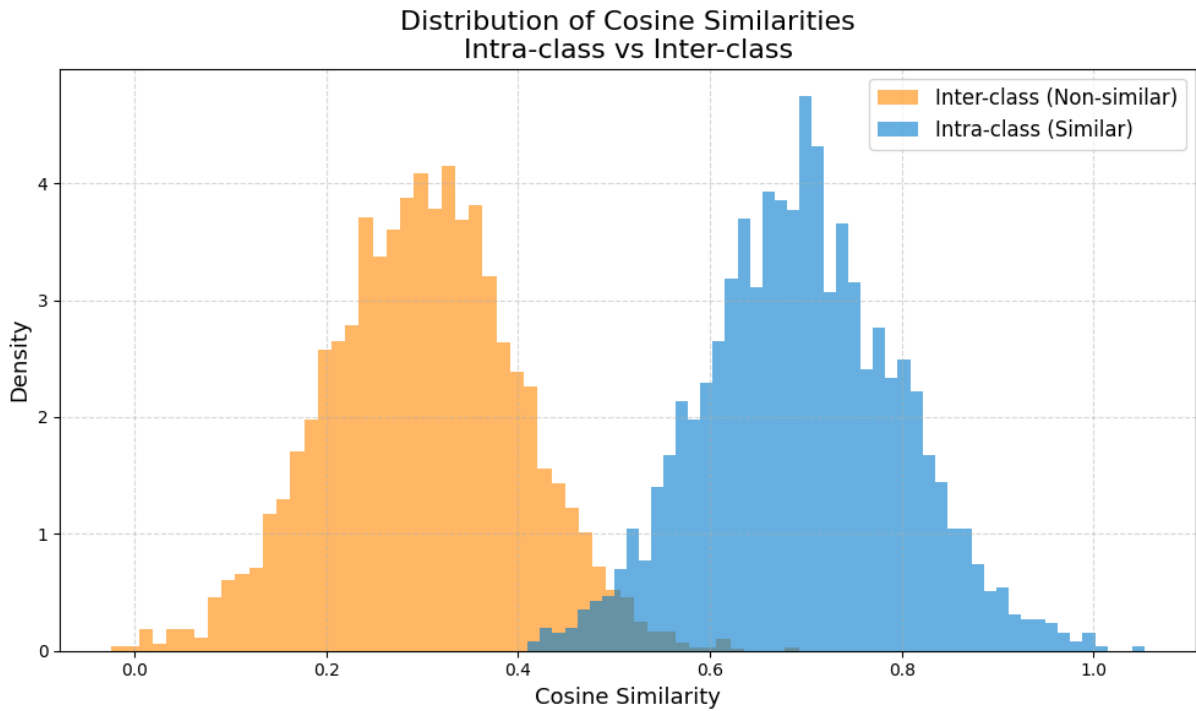


Figure 9 :Distribution of Cosine Similarities

These graphs demonstrate the model's ability to distinguish between elements. It shows excellent results. Indeed, the model assigns high scores to similar pairs, mostly ranging between 0.6 and 0.8, and low scores to dissimilar pairs, ranging between 0.2 and 0.4. The two distributions are clearly distinct with little overlap, demonstrating that the model can distinguish between similar and dissimilar items.

6. Discussion and Conclusion

6.1. Impact

This project has a particular impact on the industry. As French LLMs form the basis of several applications in important sectors such as banking and law, as well as in tasks such as translation and social media detection, this study provides a deeper understanding of language and therefore more accurate and refined services. It also paves the way for understanding local languages outside of standard registers, which are numerous in regions of the world such as Africa.

6.2. Outlook

Looking ahead, this study recommends

- Applying the model to models other than Flaubert
- Carrying out the same process with other techniques to assess relevance
- Addressing models other than informal ones, such as the local language of communities in Africa

6.3. General conclusion

Overall, this study has proven to be a success. After collecting, augmenting and cleaning the data, it was able to demonstrate how the use of DAFT alone can be insufficient. Furthermore, catastrophic forgetting also indicates that corpus size is an important factor in the process of improving LLMs and that insufficient datasets can disrupt the structure of LLMs. The main limitations encountered during the study were the collection of a large amount of data and the discovery of the use of regression head instead of AutoModel. In summary, this study significantly improved the performance of the base model, proving the relevance and success of this study.

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